

Damitri Kundu, Ph.D.

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Profiles [in LinkedIn](#) [Google Scholar](#) [GitHub](#) [ResearchGate](#)

About me

Biostatistician with experience in early-phase oncology trials, specializing in dose optimization, Bayesian modeling, and analysis of complex longitudinal and survival data. Focused on developing statistically rigorous and computationally efficient solutions for real-world problems. ([R](#), [JAGS](#), [clinical trials](#), [predictive modeling](#))

Academic Experience

My Publications

● Current Research:

- ❖ **PREDOSE : Pharmacometrically-Refined Early-phase Dose Optimization design for Oncology Study Enhancement-** Dr. Ayon Mukherjee (supervisor and project lead), Prof. Yuan Ji (*U. Chicago*), Dr. Damitri Kundu, Dr. Arunava Chakravartty and KyongJu Lee.
- ❖ **Enhancing Restricted Mean Duration of Response for Interim Monitoring in Group Sequential Rare-Disease Cancer Trials-** Dr. Ayon Mukherjee (supervisor and project lead), Dr. Micheal Grayling, Dr. Damitri Kundu and Dr. James Wason.

● Ph.D. thesis publications:

- ❖ **Kundu, D.**, Basu, S., Gogoi, M. and Das, K., 2025. A Bayesian latent class joint model for multivariate longitudinal and time-to-event data, *Journal of Applied Statistics*, pp. 1-17.
- ❖ **Kundu, D.**, Krishnan, S., Gogoi, M. and Das, K., 2024. A Bayesian quantile joint modeling for multivariate longitudinal and event-time data, *Lifetime Data Analysis*, 30(3), pp.680-699.
- ❖ **Kundu, D.**, Sarkar, P., Gogoi, M.P. and Das, K., 2023. A Bayesian joint model for multivariate longitudinal and time-to-event data with application to ALL maintenance studies. *Journal of Biopharmaceutical Statistics*, 34(1), pp.37-54.

● Publications outside thesis:

- ❖ **Kundu, D.** and Das, K., 2024. A quantile-regression approach to bivariate longitudinal joint modeling, *Journal of Statistical Research*, 58(1), pp.111-130.
- ❖ Sen, S., **Kundu, D.** and Das, K., 2023. A flexible Bayesian variable selection approach for modeling interval data. *Statistical Methods & Applications*, 33(1), pp.267-286.
- ❖ Sen, S., **Kundu, D.** and Das, K., 2023. Variable selection for categorical response: A comparative study, *Computational Statistics*, 38(2), pp.809-826.
- ❖ Kedia, P., **Kundu, D.** and Das, K., 2023. A Bayesian variable selection approach to longitudinal quantile regression. *Statistical Methods & Applications*, 32(1), pp.149-168.

● Publications from Master's thesis:

- ❖ Das, S., **Kundu, D.** and Dewanji, A., 2022. Software reliability modeling based on NHPP for error occurrence in each fault with periodic debugging schedule, *textitCommunications in Statistics-Theory and Methods*, 51(14), pp.4890-4902.

All my publications can be viewed at my ResearchGate and/or Google Scholar profiles (at the beginning of this document).

Ph.D. Experience

Indian Statistical Institute, Kolkata

Aug, 2018–Dec, 23

Key Areas: Bayesian modeling, joint modeling, survival analysis, dose optimization, R (JAGS)

- Successfully defended thesis titled **Bayesian approach to joint modeling of multivariate longitudinal and time-to-event data with applications to ALL maintenance studies** (Dec 2023).
- Developed novel Bayesian methods for joint modeling of longitudinal and survival data, with applications in clinical and biomedical studies.
- Implemented statistical models in R using JAGS for posterior inference and predictive analysis.
- Published **peer-reviewed papers** in statistical journals (see Publications section).
- Teaching Assistant for *Survey Sampling* (Coursera), managing 100+ students.
- Assisted in advanced training sessions at the *Advanced School for Statistical Learning*.

Associate Advisor - Statistics

Eli Lilly and Company, Bangalore

Sep, 24–June, 26

Key Skills: R, R Shiny App, clinical trial methodology, Bayesian modeling, dose optimization and escalation, PK/PD-informed modeling, predictive analytics, industry–academia collaboration, project grant application, external conference presentations

● Dose Optimization:

- ✓ *PREDOSE*, developing a Bayesian dose-optimization framework integrating patient-level PK/PD data within a seamless two-stage early-phase oncology trial design, aligned with **FDA Project Optimus**, under the supervision of Dr. Ayon Mukherjee.
- ✓ Developed an R Shiny web application ([PKComb-BOIN12 Simulator](#)) supporting the manuscript *PKComb-BOIN12: A Pharmacokinetics-Informed Bayesian Optimal Interval Design for Dose Optimization in Cancer Drug-Combination Trials* (Dr. Jeiqi Tu, Dr. Ruitao Lin & Dr. Ayon Mukherjee, 2026 *manuscript in preparation*).
- ✓ Developed an R Shiny web application ([OBDC-Compare](#)) and code based ([Github: OBDC-Compare](#)) supporting the manuscript *Optimal Dose Combination Selection in Oncology Trials Using Bayesian Designs: Statistical, Regulatory, and Operational Insights* (Dr. Ayon Mukherjee, Dr. Kentaro Takeda & Dr. James M.S. Wason, 2026 *manuscript in preparation*).

● Lead Statistician:

Led statistical analyses in an *early-phase oncology* study, including authoring LoA, generating TFLs from raw data, contributing to Investigator Brochure submission, and providing statistical insights to clinical, medical, and data management teams through R-based analyses.

● SMILE:

A dashboard (Study Management Insights and Learning) powered by PowerBI, with visualizations and statistical analyses generated in R, designed to predict various aspects of a clinical trial study, which contribute to decision making in the management of clinical trials:

- ✓ **Enrolment Projection:** Bayesian modeling and bootstrap sampling in R JAGS were used to project enrollment and estimate additional patients needed to meet the target with >90% probability via *bisection* and *gradient descent* optimization.
- ✓ **Evaluating Trends over Time:** Bayesian *posterior predictive distributions* were utilized to estimate and provide 95% CI for metrics AE, screening, IPD rate across various intervals of interest.
- ✓ **Early Discontinuation Projection:** Applied *bootstrap sampling* together with survival analysis to estimate discontinuation probability and characterize patient dropout patterns across cohorts.

Reduced computational runtime from **48 hours to 5 hours** across 218 studies in the SMILE dashboard by optimizing R code and leveraging high-performance libraries.

● External Presentations:

- ✓ Presented published work on **Bayesian latent class joint modeling for multivariate longitudinal and time-to-event data** (*Journal of Applied Statistics*) at *Sankhya 2025* (Novartis Hyderabad) and *ConSPIC 2025* (Trivandrum).
- ✓ Presented ongoing research on **PREDOSE: Pharmacometrically-Refined Early-Phase Dose Optimization** at *Conference on Data-driven Statistical Decision-making (CDSDM) 2026* (IIT Hyderabad, India).
- ✓ International presentations: **PREDOSE: Pharmacometrically-Refined Early-Phase Dose Optimization**, accepted at *Statisticians in the Pharmaceutical Industry (PSI) 2026* (Belfast, UK) and *Adaptive Designs and Multiple Testing Procedures (ADMTP) 2026*, Newcastle University (Newcastle, UK), June 2026.

● Industry–Academia Collaboration:

- ✓ Secured internal grant support for **PREDOSE**, enabling collaboration with Prof. Yuan Ji (*University of Chicago*).
- ✓ Conducted academic engagement sessions at Indian Statistical Institute (Kolkata) and Calcutta University.
- ✓ Contributed to proposed curriculum updates for ISI M.Stat program in *Survival Analysis* and *Clinical Trials*.

Senior Data Scientist

Intuit, Bangalore

Aug, 23–Sep, 24

Key Skill: Machine learning modeling + GenAI + prompt engineering + data-driven decision support

Project Profitability : LightGBM-based modeling; **Product-Customer Engagement** : Time series analysis, Lasso, SHAP-based explainability; **Rare-Event Anomaly Detection** : Tree-based models with feature engineering; **Customer Query to SQL (GenAI)** : Multilabel classification using BERT and prompt engineering for NER.

- ✓ **WiDS Ambassador**: Conducted a workshop on *Automated Prompt Optimization in Large Language Models*, covering RAG-based methods (KATE, Skill-kNN) with practical demonstrations and cross-model comparisons.

Associate Data Scientist

RedBus, Bangalore

Jun, 17– Aug, 2018

Workflow Prioritization : Sentiment analysis using multinomial logistic regression; **Hotel Classification** : Classification models using Random Forest; **Dynamic Pricing** : Regression-based modeling for pricing optimization; **Revenue Forecasting** : Time series models for expected marginal seat revenue.

Education

Ph.D. in Statistics

Indian Statistical Institute, Kolkata

JRF, 2018 → SRF, 2020 → Thesis defended, 2023

2018–2023

Master's Degree in Statistics (M.Stat)

Indian Statistical Institute, Kolkata

Statistics (Hons) : 73.2% [First Division]

2015–2017

Bachelor's Degree in Statistics (B.Sc)

St.Xaviers College, Kolkata

Statistics (Hons) : 69.5%[First Class]

2012–2015

High School Degree

Vivekananda Mission School

ICSE : 92% , ISC (Science and Maths) : 92%

2010–2012

Technical skills

Programming	R, RSHINY, PYTHON, PYSPARK, JAGS, BUGS, L ^A T _E X
Statistical Methods	Bayesian inference, Bayesian variable selection, survival analysis, longitudinal data analysis, joint modeling, dose optimization, PK/PD modeling, MCMC methods
Machine Learning	Time series analysis, regression (Lasso, GLM), tree-based models (Random Forest, LightGBM), anomaly detection, model explainability (SHAP, LIME)
Tools & Platforms	R Shiny, PowerBI, Jupyter Notebook, MS Excel, LLMs (Claude, ChatGPT)
Libraries	dplyr, ggplot2, tidyverse, lme4, Rjags, mvJMBayes, survival, scikit-learn, Tensor-Flow, Keras, LightGBM
RShiny Apps	(i) PKComb-BOIN12 (https://22c7ba-damitri-kundu.shinyapps.io/PKComb-BOIN12-App/), (ii) OBDC-Compare (https://22c7ba-damitri-kundu.shinyapps.io/OBDC-Compare-App/)

Languages

English First Language
Bengali Native
Hindi Third Language

Personal Interest

Painting